

Transient changes in cerebral vascular resistance during the Valsalva maneuver in man.



Measurements of cerebral spinal fluid pressure, arterial pressure, and internal carotid artery blood flow were obtained in a series of patients during a Valsalva maneuver. During straining (phase II), an 11% reduction in mean arterial pressure was associated with a 21% decrease in internal carotid flow from control values; and following release (phase IV), the 19% increase in mean arterial pressure produced a 22% increase in internal carotid artery flow. Perfusion pressure computed as the mean arterial pressure minus cerebral spinal fluid pressure and internal carotid artery blood flow were used to calculate an index of cerebral vascular resistance. The data indicate that a modest but significant decrease in vascular resistance occurred during phases II and III followed by return to control levels during phase IV. These changes in vascular resistance were not rapid enough or of sufficient magnitude to maintain constant cerebral perfusion during the Valsalva maneuver.